Recitation 8

<u>Question 1)</u>

Consider the following data set

680	669	719	699	670	710	722	663	658	634	720	690
677	669	700	718	690	681	702	696	692	690	694	660
649	675	701	721	683	735	688	763	672	698	659	704
681	679	691	683	705	746	706	649	668	672	690	724
652	720	660	695	701	724	668	698	668	660	680	739
717	727	653	637	660	693	679	682	724	642	704	695
704	652	664	702	661	720	695	670	656	718	660	648
683	723	710	680	684	705	681	748	697	703	660	722
662	644	683	695	678	674	656	667	683	691	680	685
681	715	665	676	665	675	655	659	720	675	697	663

a) Construct a stem and leaf display for these data.
b) Calculate sample mean, sample standard deviation
c) Construct a histogram for these data.

Solution

a)

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63|47
64|24899
65|223566899
66|0000001233455788899
67|0022455567899
68|00001111233333458
69|0000112345555677889
70|011223444556
71|0057889
72|000012234447
73|59
74|68
75|
76|3
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b)

$$\text{Sample Mean } \bar{x} = \frac{\sum_{i=1}^{120} x_i}{120} = \frac{82413}{120} = 686.78$$

Sample Standard Deviation

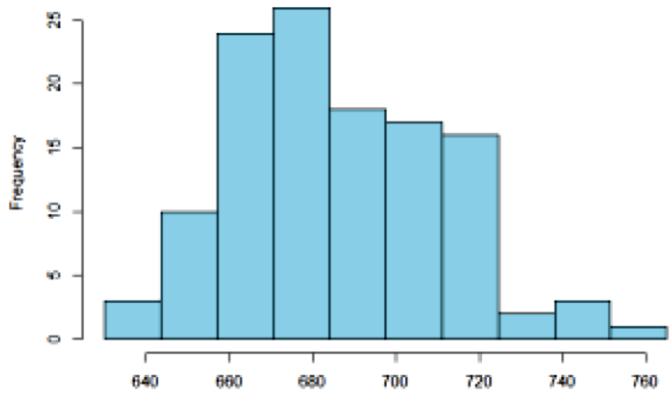
$$\sum_{i=1}^{120} x_i = 82413 \qquad \text{and} \qquad \sum_{i=1}^{120} x_i^2 = 56677591$$

$$s^2 = \frac{\sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n}}{n-1} = \frac{56677591 - \frac{(82413)^2}{120}}{120-1} = \frac{78402.925}{119}$$

$$= 658.85$$

$$s = \sqrt{658.85} = 25.67$$

Sample Median $\tilde{x} = 683.$



Question 2)

The compressive strength of concrete is normally distributed with mean 17237 kN/m^2 and standard deviation 345 kN/m^2. Find the probability that a random sample of n=5 specimens will have a sample mean diameter that falls in the interval 17230 kN/m^2 to 17305 kN/m^2

Assuming a normal distribution,

$$\mu_{\bar{X}} = 13237; \sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{345}{\sqrt{5}} = 154.289$$

$$P(17230 \le \bar{X} \le 17305) = P\left(\frac{17230-13237}{154.289} \le \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \le \frac{17305-17237}{154.289}\right)$$

$$= P(-0.045 \le Z \le 0.4407) = P(Z \le 0.44) - P(Z \le -0.045)$$

$$= 0.6700 - 0.482 = 0.188$$

Question 3)

Let $X_1, X_2, \ldots X_n$ be random sample of size n from a population with mean μ and variance σ^2 .

a) Show that $\bar{X}^2$ is biased estimator for μ^2
b) Find the amount of bias in this estimator

a)

$$E(\bar{X}^2) = \frac{1}{n^2} E\left(\sum_{i=1}^{n} X_i\right)^2 = \frac{1}{n^2}\left(V\left(\sum_{i=1}^{n} X_i\right) + \left[E\left(\sum_{i=1}^{n} X_i\right)\right]^2\right)$$

$$= \frac{1}{n^2}\left(n\sigma^2 + \left(\sum_{i=1}^{n} \mu\right)^2\right) = \frac{1}{n^2}\left(n\sigma^2 + (n\mu)^2\right)$$

$$= \frac{1}{n^2}\left(n\sigma^2 + n^2\mu^2\right) E(\bar{X}^2) = \frac{\sigma^2}{n} + \mu^2$$

Therefore, $\bar{X}^2$ is a biased estimator of μ^2.

b)

$$\text{Bias} = E(\bar{X}^2) - \mu^2 = \frac{\sigma^2}{n} + \mu^2 - \mu^2 = \frac{\sigma^2}{n}$$

Question 4

A machine produce metal rods used in a car suspension system. A random sample of 15 rods is selected and the diameter is measured. The resulting data (in mm) are as follows

8.24	8.25	8.20	8.23	8.24
8.21	8.26	8.26	8.20	8.25
8.23	8.23	8.19	8.28	8.24

a) Calculate a 95% two sided confidence bound on the mean rod diameter

b) Calculate a 95% upper sided confidence bound on the mean rod diameter

b) 95% two-sided confidence interval on mean rod diameter

For $\alpha = 0.05$ and $n = 15$, $t_{\alpha/2,n-1} = t_{0.025,14} = 2.145$

$$\bar{x} - t_{0.025,14}\left(\frac{s}{\sqrt{n}}\right) \le \mu \le \bar{x} + t_{0.025,14}\left(\frac{s}{\sqrt{n}}\right)$$

$$8.23 - 2.145\left(\frac{0.025}{\sqrt{15}}\right) \le \mu \le 8.23 + 2.145\left(\frac{0.025}{\sqrt{15}}\right)$$

$$8.216 \le \mu \le 8.244$$

95% upper confidence bound on mean rod diameter $t_{0.05,14} = 1.761$

$$\mu \le \bar{x} + t_{0.025,14}\left(\frac{s}{\sqrt{n}}\right)$$

$$\mu \le 8.23 + 1.761\left(\frac{0.025}{\sqrt{15}}\right)$$

$$\mu \le 8.241$$